Reproducing EDM on CIFAR-10: Conditioning, Classifier-Free Guidance, and FID

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Abstract

Diffusion models generate images by learning to reverse a noising process, and the design space of this reversal is large. This report reproduces key results of Elucidating the Design Space of Diffusion-Based Generative Models (EDM, Karras et al., NeurIPS 2022) on CIFAR-10: unconditional and class-conditional sampling from NVIDIA's pretrained EDM networks, the behavior of the deterministic Heun sampler examined through denoiser trajectories, and classifier-free guidance (CFG), which mixes a conditional and an unconditional denoiser estimate with a weight w. We measure label accuracy, Fréchet Inception Distance (FID), and per-class FID across $w \in \{0, 1, 1.5, 2, 3, 4\}$ at 1,000 images per setting, and we train small CFG networks ourselves on CIFAR-10 subsets and on MNIST.

Code: https://github.com/vishudhshah/edm

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1 Introduction

Diffusion models generate images by progressively transforming random noise into structured samples through a learned denoising process (Ho, Jain and Abbeel 2020). The denoising formulation behind modern samplers was unified by the score-based view (Song et al. 2021), and EDM (Karras et al. 2022) re-derives the whole pipeline — noise schedule, network preconditioning, sampler, and training weights — as a design space of separable choices, reaching state-of-the-art FID on CIFAR-10 (1.97 unconditional, 1.79 conditional) with far fewer network evaluations than prior designs.

Producing realistic images does not by itself provide control over what those images depict. Classifier-free guidance (Ho and Salimans 2022) adds that control without a separate classifier: a conditional denoiser $D(x;\sigma,c)$ and an unconditional one $D(x;\sigma)$ are mixed as $D_w = D(x;\sigma) + w(D(x;\sigma,c) - D(x;\sigma))$, with $w > 1$ extrapolating along the direction the class makes in the estimate. Guidance improves prompt adherence but trades off sample variety, which motivates measuring accuracy, FID, and per-class FID together across w.

The checkpoint work here is the base for the capstone's goal of steering pretrained diffusion models with guidance-like signals. We reproduce: (1) unconditional and conditional sampling with EDM's pretrained CIFAR-10 networks, examined inside the sampler through trajectories of the noisy image and the denoiser's estimate; (2) the CFG sweep over w with label accuracy, FID, and per-class FID at 1,000 images per setting; (3) the training of small CFG networks on 100 and 1,000 CIFAR-10 images with 10% label dropout, including the memorization check against nearest training images; (4) the same CFG construction trained from scratch on MNIST; and, as compute allows, (5) a partial run of EDM's full 55.7M-parameter training.

1.1 Data

We use CIFAR-10 (Krizhevsky 2009) (50,000 training and 10,000 test 32×32 color images in ten classes) for the pretrained-model experiments and the subset training, and MNIST (LeCun et al. 1998) MNIST (60,000 28×28 grayscale digits) for the from-scratch CFG network. All CIFAR FIDs use EDM's reference statistics of the 50,000 CIFAR-10 training images and EDM's Inception detector. The MNIST FID uses the frozen digit classifier's 128-d features, with the 10,000 real test images as the reference, since no precomputed MNIST statistics ship with the lab; these numbers are therefore comparable only across w within this report, never with published Inception-FIDs. With 1,000 samples per setting our FIDs are much higher than published 50,000-sample values, so runs are compared only with each other at equal sample counts.

2 Methods

3 Preliminary results

4 Discussion

5 Conclusion

References

Ho, Jonathan, Ajay Jain, and Pieter Abbeel. 2020. "Denoising Diffusion Probabilistic Models." In *Advances in Neural Information Processing Systems*. [Link]

Ho, Jonathan, and Tim Salimans. 2022. "Classifier-Free Diffusion Guidance." *arXiv preprint arXiv:2207.12598*. [Link]

Karras, Tero, Miika Aittala, Timo Aila, and Samuli Laine. 2022. "Elucidating the Design Space of Diffusion-Based Generative Models." In *Advances in Neural Information Processing Systems*. [Link]

Krizhevsky, Alex. 2009. "Learning Multiple Layers of Features from Tiny Images." *Technical Report, University of Toronto*. [Link]

LeCun, Yann, Léon Bottou, Yoshua Bengio, and Patrick Haffner. 1998. "Gradient-Based Learning Applied to Document Recognition." Technical Report, Proceedings of the IEEE

Song, Yang, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. 2021. "Score-Based Generative Modeling through Stochastic Differential Equations." In *International Conference on Learning Representations*. [Link]